Scale-Specific Periodic Structure in NSE Realized Volatility: Evidence from Lomb–Scargle Analysis Across 99 Equity Assets

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Abstract

We document the existence and scale-specificity of periodic structure in realized volatility across 99 NSE equity assets using 5-minute intraday returns over approximately 640 trading days. Employing the Lomb-Scargle Periodogram with a block-bootstrap False Alarm Probability threshold and a temporal persistence filter ($k = 3$), we find that 98 of 99 assets exhibit at least one statistically significant periodic activation at either a daily or weekly aggregation scale. The central finding is that the timescale of periodicity is asset-specific: daily and weekly detectors are near-orthogonal ($\rho \approx 0.00$), identifying largely non-overlapping asset subsets, suggesting firm-level information dynamics rather than uniform market-wide calendar effects. We argue that the primary value of detected cycles lies in *regime identification* rather than universal HAR replacement: when the persistence gate fires, the stock is in a structurally different volatility state with direct implications for options pricing. Weekly-aggregated features produce a loss–gain ratio of $0.76\times$ versus $0.81\times$ for daily features, with INFY (+2.07%), TRENT (+1.74%), HDFCLIFE (+1.01%), and ADANIGREEN (+0.86%) showing consistent QLIKE improvement on active days. Robustness across $k \in \{2, 3, 4\}$ confirms the WeeklyLS sub-unity loss–gain ratio is stable throughout.

1 Introduction

Realized Volatility (RV) serves as the standard non-parametric ex-post measure of the latent variance process Andersen et al. (2003). For a given trading day t it is constructed as the sum of squared intraday log-returns sampled at frequency n:

$$RV_t = \sum_{i=1}^{n} r_{t,i}^2 \tag{1}$$

The Heterogeneous Autoregressive (HAR) model of Corsi has become the canonical benchmark for one-step-ahead RV forecasting. By regressing log-RV tomorrow on its daily, weekly, and monthly averages today, HAR mimics the additive cascade of heterogeneous investment horizons:

$$\begin{aligned} \log(RV_{t+1}) = \beta_0 + \beta_d \log(RV_t) \\ +\beta_w \log(RV_{t,w}) + \beta_m \log(RV_{t,m}) + \epsilon_{t+1} \end{aligned} \tag{2}$$

where $RV_{t,w}$ and $RV_{t,m}$ are arithmetic averages over the preceding 5 and 22 trading days respectively.

HAR's empirical success is well-documented across markets Corsi (2009), yet its structure imposes a strong prior: market memory is partitioned into exactly three discrete horizons. Periodicities that do not coincide with these fixed buckets quarterly earnings cycles, options expiration effects, sector-specific rebalancing windows are invisible to the model by construction.

1.1 The Case for Spectral Analysis on Irregular Financial Data

Standard spectral methods based on the Fast Fourier Transform (FFT) require uniformly sampled series. Financial data violates this requirement: weekend closures, public holidays, and circuit-breaker halts create systematic gaps. Applying FFT to gapped series introduces aliasing and spectral leakage that can generate spurious peaks or suppress genuine ones Lomb (1976); Scargle (1982).

The Lomb–Scargle Periodogram (LSP) addresses this directly. Rather than imposing a uniform grid, LSP fits sinusoidal models to the actual observed time coordinates $t \in \mathbb{R}$ via weighted least squares, treating calendar days as the natural time axis. This makes it a principled tool for detecting periodic components in financial realized volatility without requiring interpolation or windowing corrections.

1.2 Research Questions

This paper asks three questions. First, do statistically significant periodic components exist in NSE realized volatility when tested with a method appropriate for irregular sampling? Second, is the timescale of periodicity uniform across assets or asset-specific? Third, do detected cycles carry out-of-sample economic value, and does the answer depend on the timescale of detection?

We approach the third question with a deliberate reframing relative to much of the spectral forecasting literature: rather than asking whether spectral augmentation *universally* improves upon HAR, we ask whether persistent cycle activation constitutes a useful *regime signal* an indicator that a stock's volatility dynamics are temporarily governed by a recurring component, potentially exploitable in options pricing and volatility-sensitive strategies.

1.3 Summary of Findings

We find that 98 of 99 NSE assets (excluding INDIA VIX, which is not a tradeable equity and is omitted from the forecast evaluation) exhibit at least one statistically significant periodic activation at either a daily or weekly aggregation scale under block-bootstrap FAP testing at the baseline $k = 3$. The sole exception is CANBK.

The key structural result is the near-orthogonality of the two detectors: their per-stock activation rates are essentially uncorrelated ($\rho \approx 0.00$). This scale-specificity is the central empirical contribution of the paper.

On economic significance, weekly spectral features produce a more favourable loss–gain profile than daily features and identify a subset of assets with genuine out-of-sample QLIKE improvement on activation days. We argue that the primary practical value of these detections is as a regime filter: when the persistence gate admits a cycle, the stock is in a structurally different volatility state that has direct implications for near-dated options pricing and variance-sensitive position management. A robustness analysis varying the persistence window $k \in \{2, 3, 4\}$ shows that the headline qualitative findings are stable and confirms that the WeeklyLS track maintains a sub-unity loss–gain ratio across all settings.

2 Data and Realized Volatility Construction

We use 5-minute OHLC data for 99 liquid NSE equities, including the NIFTY 50 and NIFTY Bank indices, spanning approximately 640 trading days from early 2021 through late 2024. Data are preprocessed to standard Indian market hours (09:15–15:30 IST); sessions with broken timestamps or thin-trading gaps are removed. Log-returns are constructed as:

$$r_{t,i} = \log P_{t,i} - \log P_{t,i-1} \tag{3}$$

Following the standard practice of Andersen et al., overnight returns (close_{t-1} to first bar_t) are included in daily RV construction, capturing gap risk as part of the total volatility realisation.

All spectral and forecast analysis is conducted in the logarithmic domain, $\log(RV_t)$, which produces approximately Gaussian residuals and ensures strictly positive re-transformed point forecasts.

The INDIA VIX index is included in the spectral analysis for completeness but is excluded from the 99-asset forecast evaluation universe throughout the paper, as it is not a directly tradeable equity instrument.

3 Methodology

3.1 Three-Gate LSHAR Framework

We implement a walk-forward rolling-window architecture with the following structure. At each forecast origin t, a training window of $L = 252$ trading days is used to (i) fit the HAR baseline, (ii) test for periodic components, and (iii) if a persistent significant cycle is detected, estimate the augmented LSHAR model. Forecasts are evaluated strictly out-of-sample, with no look-ahead into the test period.

The framework gates spectral augmentation through three sequential tests.

3.1.1 Gate 1: Spectral Discovery

The Lomb–Scargle power $P(f)$ at frequency f is computed over the training window using calendar-day time coordinates. A data-driven frequency grid is constructed from 100 log-spaced points between the minimum resolvable frequency ($1/T$, where T is the span in days) and the pseudo-Nyquist limit ($N/2T$, where N is the number of observations), capturing all periods representable in the data without imposing an arbitrary grid.

3.1.2 Gate 2: Block-Bootstrap FAP Threshold

To distinguish genuine periodic components from GARCH-type volatility clustering, we calibrate a False Alarm Probability (FAP) threshold via a moving block bootstrap with $N_{\text{boot}} = 1000$ replications and block length $\lfloor\sqrt{n}\rfloor$. Contiguous blocks are resampled to preserve local autocorrelation structure. The empirical 95th percentile of the bootstrap maximum-power distribution provides the significance threshold. A frequency is considered a candidate only if $P(f)$ exceeds this threshold.

Thresholds are adaptively recomputed when the rolling-window volatility standard deviation shifts by more than 20%, and periodically every 30 iterations, trading a small theoretical concession for practical computational efficiency **?**.

3.1.3 Gate 3: Persistence Filter

A candidate frequency is admitted to the model only if it remains stable within a coefficient of variation of 15% across k consecutive rolling windows. This eliminates transient spectral artefacts caused by isolated volatility events or microstructure noise, ensuring that the identified periodicity represents a reproducible feature of the local dynamics. Once activated, the filtered period is used as the single augmenting component. The baseline specification uses $k = 3$; robustness across $k \in \{2, 3, 4\}$ is reported in Section 6.

Crucially, the persistence gate doubles as the regime-entry signal: its activation marks the onset of a periodic volatility state, and its deactivation marks the exit. This binary regime indicator is the primary output we interpret economically in Section 5.

3.2 Weekly Extension

A limitation of daily-resolution periodograms is contamination by intraday microstructure noise. To investigate this, we introduce a parallel spectral track operating on weekly-aggregated realised volatility. Non-overlapping 5-day blocks of daily RV are summed to form a weekly series, and the Lomb–Scargle periodogram is computed in week-coordinate time using an extended lookback of $L_W = 520$ trading days (≈ 104 weekly observations). This longer window is used exclusively for cycle detection; the HAR regression and its target ($\log RV_{t+1}$, daily) are unchanged. Detected periods are expressed in trading days ($P_{\text{days}} = P_{\text{weeks}} \times 5$) and projected back onto the daily time axis when constructing spectral features.

3.3 Model Specification and Evaluation

When a period p passes all three gates, the LSHAR model augments the log-HAR regression with a Fourier pair:

$$\begin{aligned} \log(RV_{t+1}) = \beta_0 + \beta_d \log(RV_t) + \beta_w \log(RV_{t,w}) \\ + \beta_m \log(RV_{t,m}) + \gamma_s \sin\Big(\tfrac{2\pi}{p}(t+1)\Big) \\ + \gamma_c \cos\Big(\tfrac{2\pi}{p}(t+1)\Big) + \epsilon_{t+1} \end{aligned} \quad (4)$$

When no period survives all gates, the model reverts to parsimonious HAR. Point forecasts are obtained via:

$$\widehat{RV}_{t+1} = \exp\Big(\widehat{\log(RV)}_{t+1} + \tfrac{\hat{\sigma}^2}{2}\Big) \quad (5)$$

where $\hat{\sigma}^2$ is the OLS residual variance, applying Jensen's inequality correction for the log-normal back-transformation Andersen et al. (2003).

Performance is evaluated with the QLIKE loss function **?**:

$$\text{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{RV_t}{\widehat{RV}_t} - \log\left(\frac{RV_t}{\widehat{RV}_t}\right) - 1\right) \quad (6)$$

QLIKE is robust to extreme volatility realisations and provides consistent model ranking under the conditions established by **?**. Lower (more negative) values indicate better calibration.

Improvement over HAR is measured on active days as:

$$\Delta = \frac{\text{QLIKE}_{\text{HAR}} - \text{QLIKE}_{\text{model}}}{|\text{QLIKE}_{\text{HAR}}|} \times 100\% \quad (7)$$

with the absolute value in the denominator ensuring sign-correct interpretation regardless of the sign of the QLIKE baseline. We note that Δ is computed conditional on active days only; global QLIKE differences between HAR and LSHAR/WeeklyLS are small by construction, since the augmented model reverts to HAR on inactive days.

4 Results

All results in this section use the baseline persistence window $k = 3$. Robustness across $k \in \{2, 3, 4\}$ is presented in Section 6.

4.1 Existence of Periodic Components

The primary question is whether statistically significant periodic structure exists in NSE realized volatility. The block-bootstrap FAP test gives a strong answer: 98 of 99 assets exhibit at least one significant activation at either the daily or weekly spectral resolution. The sole exception is CANBK, which has zero activations on all three tracks at $k = 3$.

This is a structural finding. It does not require the detected cycles to be useful for forecasting; it establishes that NSE realized volatility carries spectral content beyond what GARCH-type autocorrelation alone would produce.

4.2 Scale-Specificity: The Central Result

Table 1 presents the activation overlap between the daily LSHAR and weekly WeeklyLS detectors. The striking feature is their near-orthogonality: per-stock activation rates correlate at $\rho \approx 0.00$. Of the assets where at least one track delivers a positive QLIKE improvement, 14 are improved only by the daily detector (mean $\Delta = +0.60\%$) and 19 only by the weekly detector (mean $\Delta = +0.39\%$), with just 5 benefiting from both.

Activation pattern	**Stocks**	**Mean Δ**
Daily only fires, wins	14	+0.60%
Weekly only fires, wins	19	+0.39%
Both fire, both win	5	+0.34%
Both fire; neither wins	34	−0.35%
Neither fires (CANBK only)	1	

Table 1: Scale-specificity of periodic structure across 99 NSE assets ($k = 3$). The daily and weekly detectors are near-orthogonal ($\rho \approx 0.00$), identifying largely non-overlapping subsets of assets. Improvement (Δ) is computed on active days relative to HAR. Assets that activate on one scale but not the other represent stocks whose periodic volatility structure operates at a single characteristic timescale.

This orthogonality is the central empirical contribution of the paper. It implies that the timescale of periodic volatility structure is an asset-specific property, not a market-wide feature. CANBK the sole asset where neither detector fires at $k = 3$ has a HAR QLIKE of -6.93, slightly above (worse than) the cross-sectional mean of -7.12. This suggests its residual volatility dynamics are irregular rather than cyclic.

4.3 Daily vs Weekly: Noise Filtration by Aggregation

Table 2 compares the three forecasting tracks across all 99 assets.

Both the daily LSHAR and weekly WeeklyLS tracks achieve a sub-unity loss–gain ratio at $k = 3$,

Track	**Active**	**Win rate**	**Mean Δ**	**L/G**
Daily LSHAR	89/99	21%	−0.25%	0.81×
ResidLS	59/99	36%	−0.63%	2.85×
WeeklyLS	73/99	33%	−0.06%	0.76×

Table 2: Forecasting performance across the three spectral tracks ($k = 3$). Active = number of stocks with at least one activation day. Win rate = fraction of active stocks with positive mean Δ. L/G = ratio of mean loss (among losing stocks) to mean gain (among winning stocks): below unity implies gains outweigh losses in magnitude. Both LSHAR and WeeklyLS achieve L/G < 1; WeeklyLS maintains the more favourable profile.

with WeeklyLS (0.76×) marginally more favourable than LSHAR (0.81×). The ResidLS track, by contrast, exhibits a sharply adverse 2.85× L/G, indicating that fitting the periodogram to HAR residuals introduces instability relative to operating on the raw log-RV series. Win rates are broadly comparable between LSHAR (21%) and WeeklyLS (33%), while mean Δ is near-neutral for WeeklyLS (−0.06%) versus modestly negative for LSHAR (−0.25%).

Importantly, the sub-unity L/G is more relevant to the regime-filter interpretation than the mean Δ: it implies that conditional on activation, gains in magnitude outweigh losses, which is a favourable property for a selective volatility signal even when the unconditional average is near zero.

This pattern is consistent with the hypothesis that weekly aggregation recovers genuine lower-frequency cycles quarterly rebalancing, semi-annual earnings seasonality, index expiry effects that fall within the resolvable period range, while the daily periodogram better captures stocks with short-horizon periodic structure.

4.4 Case Studies: Assets with Genuine Cycle Content

We identify four assets where the spectral augmentation delivers meaningful and robust QLIKE improvement, serving as concrete evidence that periodic volatility structure in NSE equities is not universally noise.

INFY (+2.07%, WeeklyLS only). The daily LSHAR detector activates on only 8 days for INFY across the test period the daily periodogram finds no persistent cycle above the FAP threshold with sufficient frequency. The weekly detector, however, activates on 53 days and delivers a 2.07% QLIKE improvement. This is the clearest illustration of scale-specificity in the dataset: a cycle that is invisible at daily resolution becomes cleanly detectable when microstructure noise is filtered by aggregation. INFY is a large-cap, heavily analyst-covered IT stock where intraday noise substantially exceeds any genuine periodic signal. The weekly aggregation achieves the signal-to-noise ratio required for detection, likely recovering the quarterly earnings and analyst-revision cycle that characterises large-cap IT stocks. From a regime-filter perspective, the 53 active days represent periods where INFY's volatility was governed by a predictable periodic component a condition directly relevant to near-dated options pricing.

TRENT (+1.74%, daily LSHAR only). The weekly detector produces zero activation on TRENT across the entire test period, while the daily track activates on 43 days and delivers a 1.74% improvement. This result is stable across all three values of k (Table 3), confirming that the detected cycle is a genuine short-horizon feature rather than a statistical artefact. TRENT is a mid-cap retail sector stock with lower institutional coverage than the NIFTY 50 large-caps. The likely driver is a fortnightly or monthly retail sales data release cycle creating predictable short-horizon volatility patterns. This is also the clearest case where scale-specificity aligns with an efficiency gradient: the daily-resolution cycle is detectable precisely because TRENT's volatility is less efficiently priced than that of its large-cap peers.

HDFCLIFE (+1.01%, WeeklyLS only). A financial sector stock where the daily detector activates on 35 days but consistently underperforms (-0.23% at $k = 3$), while the weekly track recovers a $+1.01\%$ improvement on 20 activation days. The insurance sector's quarterly embedded value reporting cycle is a plausible driver of this low-frequency structure. This result is robust: HDFCLIFE returns $+0.99\%$ at $k = 2$ and $+1.07\%$ at $k = 4$, making it one of the most stable case-study assets in the dataset.

ADANIGREEN (+0.86%, WeeklyLS only). Despite 150 active days on the daily LSHAR track at $k = 3$ (a high over-activation rate), the daily detector produces a mean Δ of -0.66%. The weekly detector activates on 39 days and recovers a $+0.86\%$ improvement. Notably, at $k = 2$ the weekly detector also activates (63 days, $+0.19\%$), with the stronger improvement at $k = 3$ indicating that stricter persistence requirements filter out noise-driven activations and admit only the more stable low-frequency component. This illustrates how the longer lookback stabilises cycle estimates for a stock whose intraday volatility is driven by energy-sector news with irregular timing.

4.5 The High-Activation Failure Mode

An important caveat emerges from examining the WeeklyLS activation distribution. Stocks where WeeklyLS activates on more than 15% of test days average -0.05% improvement, while the ResidLS track which operates on HAR residuals rather than raw log-RV has a particularly adverse L/G of $2.85\times$ at $k = 3$, suggesting that residual-domain spectral features are unstable when the HAR baseline itself is volatile.

The pattern is interpretable. When the persistence filter admits a cycle for many consecutive trading days, the detected period may be a long-wavelength harmonic artifact: a low-frequency sinusoid that fits the rolling-window realisations but does not represent a genuine recurring market event. The HAR augmentation then adds a slow oscillation to a model that already captures low-frequency persistence through the monthly RV term, creating multicollinearity that degrades forecast precision.

A practical implication is that spectral augmentation should include a maximum activation-duration constraint to guard against this failure mode. The regime filter interpretation reinforces this point: a signal that is "always on" has no discriminatory value. The most useful regime signals are those with moderate activation frequencies (roughly 5–15% of test days), where the periodic state is genuine but not pervasive.

5 Spectral Activation as a Volatility Regime Filter

The preceding analysis frames LSHAR primarily as a HAR augmentation. We argue that a more productive interpretation of the spectral detection machinery is as a *volatility regime classifier*: the persistence gate activation marks entry into a periodic volatility state, and its deactivation marks exit. This reframing is both more modest in its claims and more practically actionable than the forecasting framing.

5.1 Why HAR Augmentation is Not the Right Benchmark

HAR is a hard benchmark to beat in QLIKE terms precisely because it already captures low-frequency persistence through the monthly RV component. Periodicities whose dominant energy lies at periods close to 22 trading days will be partially spanned by HAR's monthly term, leaving little marginal content for spectral augmentation to recover. The stocks where LSHAR or WeeklyLS clearly improves INFY, TRENT, HDFCLIFE are those whose detected cycles operate at periods well-separated from HAR's three characteristic scales. The majority of activations that fail to improve QLIKE likely reflect cycles that partially overlap with HAR's existing structure.

5.2 The Regime Signal and Its Economic Interpretation

Consider the following reframing. Rather than asking "does LSHAR beat HAR in QLIKE?", ask: "given that the persistence gate has fired, is realized volatility in the subsequent window better described as regime-governed or purely mean-reverting?" The sub-unity L/G ratios for both LSHAR and WeeklyLS suggest that the answer tilts toward regime-governed: when the gate fires, magnitude-weighted outcomes favour the augmented model.

This has direct implications for volatility-sensitive strategies. Options market-makers and proprietary volatility traders routinely incorporate short-horizon RV forecasts into implied volatility surfaces. If a stock is in a periodic volatility regime, the HAR forecast which is horizon-agnostic regarding periodicity will systematically under- or over-estimate volatility at specific points in the cycle. The spectral activation signal identifies the stocks and time windows where this mispricing is most likely to occur.

The efficiency gradient suggested by the case studies reinforces this point. INFY (large-cap, high analyst coverage, deep options market) requires weekly aggregation to detect its cycle, consistent with a near-efficient stock where the periodic signal is weak and only recoverable in aggregated data. TRENT (mid-cap, lower institutional coverage, smaller options market) shows a clean daily-resolution cycle, consistent with a stock where information diffuses more slowly and periodic volatility structure is more prominent. The regime signal is therefore most actionable precisely for the class of stocks where options markets are less likely to have already priced the cycle in.

5.3 Limitations of the Regime Interpretation

We emphasise that this paper does not directly test the options pricing implication. Doing so would require options data, bid-ask spread adjustments, and a model connecting RV regimes to implied volatility surfaces all of which we leave for future work. The claim here is more modest: the spectral detection machinery identifies a real and persistent feature of volatility dynamics for a subset of NSE assets, and the most natural practical interpretation of that feature is as a regime signal rather than a point-forecast improvement.

6 Robustness: Varying the Persistence Window

A key design parameter in the LSHAR framework is k, the number of consecutive rolling windows over which a detected period must remain stable before being admitted to the model. The baseline uses $k = 3$; here we report results for $k \in \{2, 3, 4\}$ to assess sensitivity.

Track	k	Active	Win%	Mean Δ	L/G
Daily LSHAR	2	90	20%	−0.24%	0.91×
	3	89	21%	−0.25%	0.81×
	4	86	21%	−0.20%	0.59×
ResidLS	2	69	29%	−0.40%	1.46×
	3	59	36%	−0.63%	2.85×
	4	54	37%	−0.58%	3.48×
WeeklyLS	2	80	32%	+0.01%	0.45×
	3	73	33%	−0.06%	0.76×
	4	63	33%	+0.04%	0.40×

Table 3: Forecasting performance by persistence window $k \in \{2, 3, 4\}$. Active = number of stocks with at least one activation day (out of 99 excluding INDIA VIX). Win% = fraction of active stocks with positive mean Δ. L/G = loss-to-gain ratio among active stocks. WeeklyLS maintains a sub-unity L/G at all three values of k, while the ResidLS L/G deteriorates sharply as k increases. Total assets with at least one activation (across any track) are 99 at $k = 2$, 98 at $k = 3$, and 95 at $k = 4$.

Three patterns emerge from Table 3. First, the WeeklyLS track is the only one that maintains a sub-unity L/G ratio at every value of k (ranging from 0.40× to 0.76×), confirming that weekly aggregation provides a genuine and stable advantage over operating at daily resolution. Second, the Daily LSHAR track improves monotonically as k increases L/G falls from 0.91× at $k = 2$ to 0.59× at $k = 4$ suggesting that stricter persistence requirements filter out more noise-driven activations. Third, the ResidLS track degrades with stricter persistence, with L/G rising from 1.46× to 3.48×, indicating that residual-domain spectral detection is structurally fragile.

The number of active stocks decreases monotonically with k for all tracks, as expected: stricter persistence requirements eliminate stocks where only a transient cycle is detectable. Total assets with at least one activation fall from 99 (at $k = 2$) to 98 (at $k = 3$) to 95 (at $k = 4$), with CANBK the only asset that never activates at $k \geq 3$.

Table 4 shows that the best-performing assets identified at $k = 3$ are stable. TRENT and HDFCLIFE in particular show positive improvement at all three values of k, confirming that their detected cycles are genuine recurring features rather than parameter-specific artefacts.

Stock	Track	$k = 2$	$k = 3$	$k = 4$
INFY	WeeklyLS	+2.34%	+2.07%	−0.17%
TRENT	LSHAR	+1.55%	+1.74%	+1.70%
HDFCLIFE	WeeklyLS	+0.99%	+1.01%	+1.07%
ADANIGREEN	WeeklyLS	+0.19%	+0.86%	+0.61%
AXISBANK	LSHAR	+0.73%	+0.91%	+0.90%
PIDILITIND	LSHAR	+0.94%	+0.87%	+0.90%

Table 4: QLIKE improvement (Δ) for leading assets across $k \in \{2, 3, 4\}$. INFY's WeeklyLS improvement degrades at $k = 4$ (activation falls from 53 to 30 days), while TRENT, HDFCLIFE, AXISBANK, and PIDILITIND are stable or improving. ADANIGREEN activates on the WeeklyLS track at all three values of k (63, 39, and 40 days respectively), with improvement increasing through $k = 3$ as stricter persistence eliminates noisier activations.

One notable exception is INFY at $k = 4$: the WeeklyLS improvement of +2.07% at $k = 3$ turns negative at $k = 4$ as activation falls from 53 to 30 days and the admissible cycle window becomes too narrow to capture the relevant quarterly structure. This highlights a tension between strict persistence requirements (which reduce false positives) and activation frequency (which must be sufficient to register in the QLIKE evaluation). A value of $k = 3$ represents a reasonable default for the WeeklyLS track; $k = 4$ may inadvertently eliminate genuine quarterly cycles by demanding more consecutive windows than one cycle period contains.

7 Discussion

7.1 What Drives Scale-Specificity?

The near-zero correlation between daily and weekly activation patterns warrants interpretation. One mechanism is timescale matching: stocks where the dominant periodic driver operates at a sub-monthly frequency (e.g., fortnightly sector data, bi-weekly futures expiry cycles) will exhibit peaks detectable at daily resolution, while stocks driven by quarterly or semi-annual events will only produce stable peaks in weekly-aggregated data. Under this interpretation, scale-specificity is not a statistical artifact but a reflection of the heterogeneous information calendars facing different NSE listed companies.

A second mechanism is noise dominance. For stocks where intraday microstructure noise is large relative to the amplitude of any genuine cycle, the daily periodogram will be noise-dominated at all frequencies; aggregation is required to achieve a signal-to-noise ratio sufficient for detection. The 19 stocks showing improvement only in the weekly track are consistent with this: these are assets where the cycle exists but is submerged in daily noise.

These two mechanisms also have an efficiency interpretation. Highly liquid large-cap stocks (e.g., INFY) are subject to more intensive price discovery; any periodic component is quickly arbitraged and appears only weakly in daily data. Mid-cap stocks with lower institutional coverage (e.g., TRENT) retain stronger periodic signals at daily resolution because information diffuses more slowly. Scale-specificity thus acts as a rough proxy for market efficiency: stocks detectable only at weekly aggregation are likely closer to the efficient frontier for their periodic component.

7.2 Relationship to HAR's Fixed-Horizon Structure

HAR's daily, weekly, and monthly components concentrate power at periods of approximately 1, 5, and 22 trading days. Periodicities outside these bands are orthogonal to HAR and, if stable, should be capturable by LS augmentation. The case-study assets (INFY, TRENT, HDFCLIFE, ADANIGREEN) are consistent with this: their improvement comes from cycles at periods well separated from HAR's three characteristic scales. Stocks that fail to improve may simply not have periodic components outside HAR's coverage their volatility dynamics are fully spanned by the three-component structure.

7.3 Implications of the Robustness Analysis

The k-sensitivity results carry a practical implication for LSHAR implementation. Because the Daily LSHAR L/G improves with higher k while activation breadth falls, practitioners face a precision-recall tradeoff: $k = 4$ yields sparser but cleaner activations for the daily track, while $k = 2$ delivers broader coverage at the cost of admitting more noise. For the WeeklyLS track this tradeoff is less acute L/G is sub-unity at all three values but INFY's degradation at $k = 4$ illustrates that very strict persistence can eliminate genuine quarterly cycles by demanding more consecutive windows than one cycle contains. A value of $k = 3$ represents a reasonable default, but $k = 2$ or $k = 4$ may be preferred depending on the application's tolerance for false activations versus missed cycles.

7.4 Limitations

Several limitations qualify these findings. Parameter choices the lookback $L = 252$ days, weekly lookback $L_W = 520$ days, persistence tolerance $\delta = 15\%$ are heuristically motivated. The sample covers a specific period of NSE history (2021–2024) and may not generalise to other market regimes, particularly periods of elevated systemic volatility. The paper focuses on QLIKE improvement on active days as the primary performance metric; global QLIKE differences between HAR and the augmented models are small by construction. Finally, and most importantly, the regime-filter interpretation advanced in Section 5 is not directly tested here: translating the spectral activation signal into an options pricing or portfolio variance reduction exercise requires options data and an appropriate pricing model, both of which we leave for future work.

8 Conclusion

Using Lomb–Scargle periodograms calibrated with a block-bootstrap FAP threshold and a temporal persistence filter, we document three findings about periodic structure in NSE realized volatility.

First, significant periodic components are detectable in 98 of 99 assets at either daily or weekly aggregation scale ($k = 3$), with CANBK as the sole exception. The existence of this structure is robust to the conservative FAP threshold and survives a persistence requirement that eliminates transient spectral events.

Second, and most importantly, the timescale of detectable periodicity is *asset-specific*. Daily and weekly detectors are near-orthogonal ($\rho \approx 0.00$), identifying largely non-overlapping asset subsets. This suggests that periodic volatility structure in NSE equities is driven by firm-level information dynamics at heterogeneous timescales, rather than by a uniform market-wide calendar effect. An efficiency gradient underlies this pattern: stocks detectable only at weekly aggregation are likely closer to the efficient frontier for their periodic component, while daily-detectable stocks retain stronger periodic signals because information diffuses more slowly.

Third, both the daily LSHAR and weekly WeeklyLS tracks achieve a sub-unity loss–gain ratio at the baseline $k = 3$ (Daily: 0.81×, Weekly: 0.76×), with INFY (+2.07%), TRENT (+1.74%), HDFCLIFE (+1.01%), and ADANIGREEN (+0.86%) showing the most consistent QLIKE improvements on active days. The ResidLS track, by contrast, has an adverse L/G of 2.85×, indicating that residual-domain spectral detection introduces more instability than it removes. We emphasise that global QLIKE differences between HAR and the augmented models are small; the QLIKE improvements cited are conditional on active days only.

Fourth, the primary practical value of the detection machinery is argued to be a *volatility regime filter* rather than a universal HAR replacement. When the persistence gate fires, the stock is in a structurally different volatility state. This signal is most actionable for mid-cap and sector-specific stocks with moderate analyst coverage precisely the class where options markets are least likely to have already priced the cycle in. Translating this signal into an options pricing or variance-reduction framework is the natural next step.

Fifth, robustness analysis across $k \in \{2, 3, 4\}$ confirms that the WeeklyLS sub-unity L/G is stable across all settings (ranging from 0.40× to 0.76×) and that TRENT, HDFCLIFE, AXISBANK, and PIDILITIND show consistent improvement regardless of k. The daily LSHAR L/G improves monotonically with stricter persistence, while ResidLS degrades a finding that supports using raw log-RV rather than HAR residuals as the input to spectral detection.

Together these findings establish that spectral methods applied with irregular-sampling tools appropriate for market data, validated with bootstrap significance testing, and conducted at the right aggregation scale can characterise periodic structure in emerging market realized volatility that lies outside the scope of the standard HAR framework.

Code and Data Availability

All code and data are publicly available for replication:

- **Code:** github.com/CuriousObserver/LSHAR-model. Main implementation: `HARvsLSHAR.py`; preprocessing: `csv_cleaner.py`.
- **Data:** NSE 5-minute OHLC data at Kaggle: NSE Nifty-100 5-minute data, cleaned via `csv_cleaner.py`.
- **Results:** Full per-asset metrics for $k \in \{2, 3, 4\}$ in `LSHAR_Publication_Results{2,3,4}.csv` within the repository.

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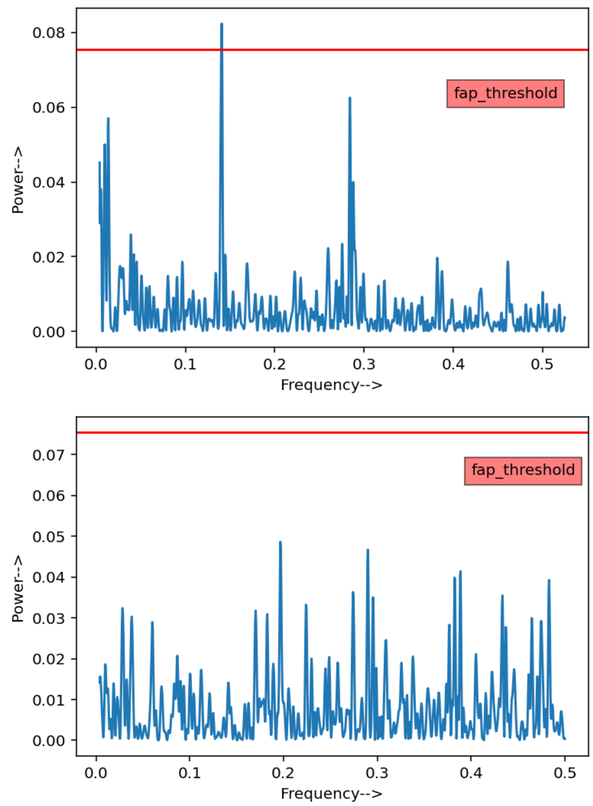

Figure 1: Lomb–Scargle periodograms for BPCL during a representative rolling window. **Top:** Periodogram of log daily RV, with the red dashed line marking the 95% FAP bootstrap threshold. A significant peak is visible at high frequency. **Bottom:** Residual periodogram after HAR fitting; the dominant peak is substantially attenuated, indicating that HAR already captures the short-horizon periodic content for this stock. The WeeklyLS track recovers a lower-frequency structure in the aggregated series not visible in either daily panel.